Annex I. Debt Sustainability Analysis

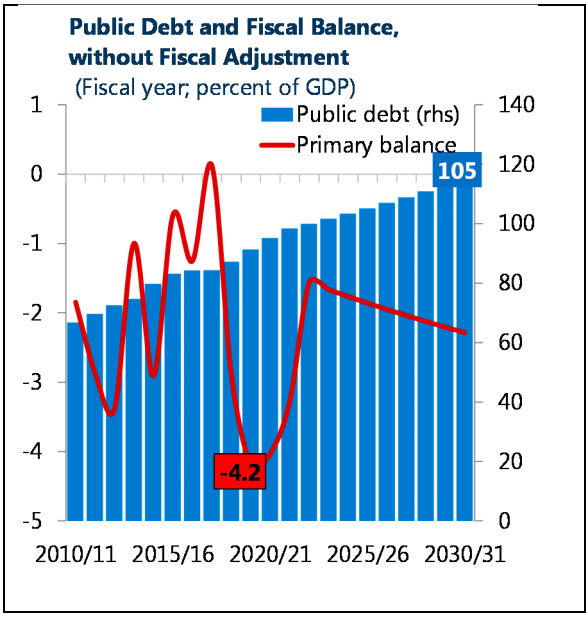
Dominica continues to be at high risk of debt distress. In the wake of tropical storm Erika, rehabilitation and reconstruction activities will entail significant fiscal costs in the short term, which would worsen the debt outlook in the absence of significant consolidation. Nevertheless, the authorities have expressed commitment to a significant fiscal adjustment that would put debt on a sustainable path and close to the ECCU regional debt target over the medium term. This constitutes the baseline scenario for this DSA. A customized scenario assuming no consolidation, complemented by a historical scenario reflecting a weak fiscal track record, is used to inform the external risk rating, given the challenging medium-term fiscal consolidation under the baseline. Should the planned fiscal consolidation take hold and the debt outlook move firmly along the sustainable path projected under the baseline scenario, an improvement of the risk rating could be considered in a future assessment.

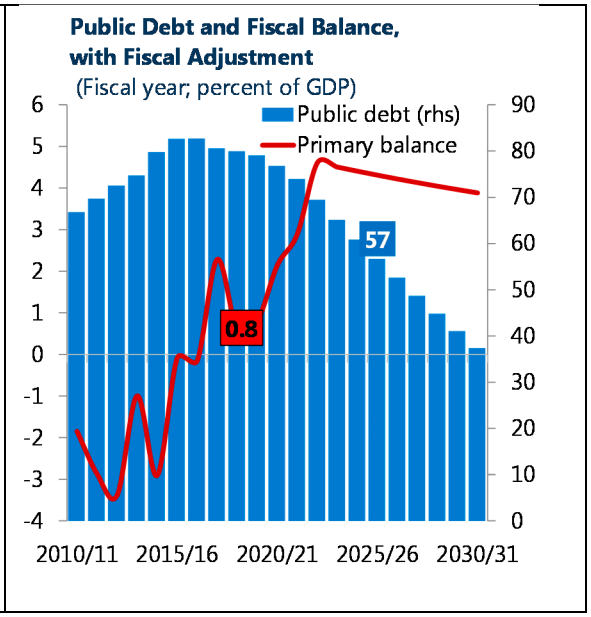
1. **Dominica's public debt has been increasing steadily since 2010, reflecting the weakening fiscal position.** As of end-FY 2014/15, the stock of central government's direct and guaranteed debt (henceforth public debt) is estimated to be around 80 percent of GDP. Over 70 percent of the debt is external, owed mainly to multilateral creditors and the remaining is held domestically, mostly by commercial banks and other financial institutions. The guaranteed debt of state owned enterprises is about 4 percent of GDP. The composition of debt has increasingly shifted towards bilateral lenders (mostly concessional), due to the growing importance of China and Venezuela, who together hold more than 11 percent of the total public debt.

2. **Under the baseline scenario, supported by the authorities' commitment in the LOI, debt is assessed to be sustainable**. This scenario envisages a gradual fiscal consolidation over the next five years, supported by comprehensive reforms to improve revenue collection, broaden the tax base, and contain current expenditures. This would significantly improve the debt outlook, setting it on a gradually declining path and allowing the authorities to meet the public debt target by FY 2025/26 (Text figure).

3. **The authorities agreed to building fiscal buffers in light of frequent natural disasters.** Dominica is subject to frequent natural disasters and the authorities are enhancing their efforts to build resilience against future storms. In the aftermath of the tropical storm Erika, the authorities have decided to build a fiscal cushion of about 1.5 percent of GDP every year beginning in FY 2018/19, which may serve as a contingency fund to provide resilience to fiscal sustainability. The baseline scenario, which envisages a primary surplus of around 4 percent of GDP over the medium-term, includes the buffer to cover future disaster related expenses.

4. **Nevertheless, significant risks of debt distress remain.** Failure to implement the necessary fiscal measures, lower-than expected donor grants, including to finance reconstruction needs, and future recurrence of natural disasters could cause debt-to-GDP ratio to breach its threshold over the medium term. This was the case even before Tropical Storm Erika hit Dominica, as flagged in the last Article IV consultation; the reconstruction and rehabilitation costs (amounting to 96 percent of GDP for the country as a whole, of which about 2/3 would be covered by the public sector) only reinforce this view. The impact of insufficient fiscal consolidation is illustrated in the customized scenario, which assumes no additional fiscal measures. [1] The debt path under the historical scenario leads to similar results reflecting a weak track record of fiscal policies.





5. **As a result, the public and publicly guaranteed (PPG) external debt risk rating remains high.** This assessment is based on the risks reflected both in the customized and historical scenarios. Given the challenges associated to the required fiscal adjustment, the customized scenario (which assumes no fiscal consolidation) is attached significant weight in this assessment. Under this scenario, the debt path takes an increasing trajectory and eventually breaches one solvency threshold (Figure A1). The historical scenario, which assumes that growth, the primary deficit, and nominal interest rates remain at historical averages, illustrates the risk inherent to the baseline

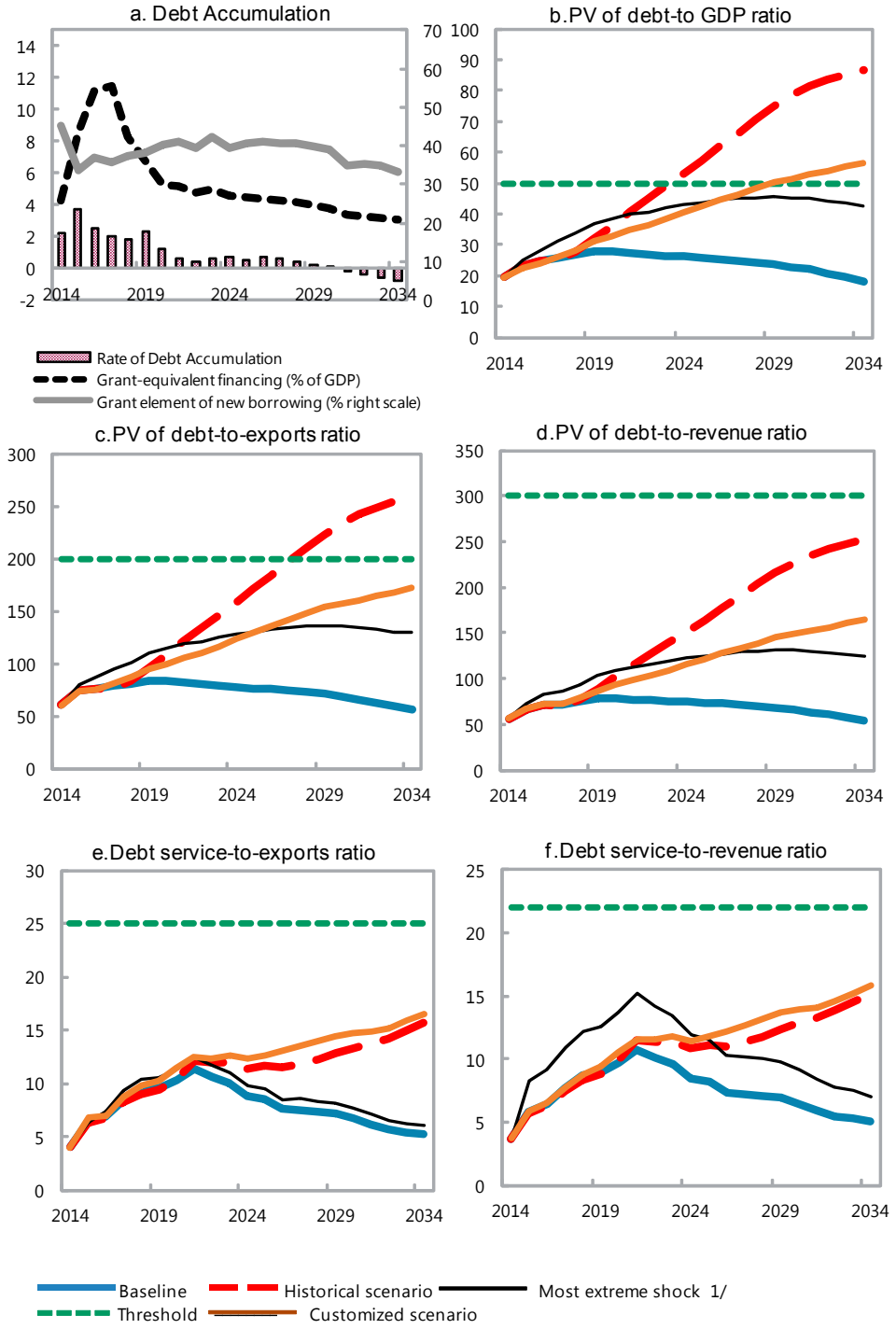
[1] This customized scenario is slightly more favorable than the passive scenario considered in the last Article IV Consultation including because of data updates and because it assumes larger grants in the medium term, which helps to partly offset the negative impact of storm Erika.

scenario, with all indicators on an increasing trajectory and two of them breaching their respective thresholds.

Conclusion

6. **The risk rating assigned to PPG external debt is high.** Under a scenario without fiscal consolidation (customized scenario) debt would exceed a solvency threshold in the medium term. Given the challenges associated to fiscal consolidation in the wake of storm Erika, this scenario is attached significant weight in the risk assessment. Nevertheless, the authorities have expressed commitment to a strong fiscal adjustment that would put public debt on a sustainable trajectory. As the planned fiscal consolidation takes hold, further increasing confidence in the baseline scenario, an upgrade of the risk rating could be considered in a future assessment.

Figure A1. Dominica: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2014-34 (Baseline Scenario) 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Customized scenario shock; in c. to a Customized scenario shock; in d. to a Customized scenario shock; in e. to a Customized scenario shock and in figure f. to a Customized scenario shock.

Table A1. Dominica: External Debt Sustainability Framework, Baseline Scenario, 2011-34 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**76.8**	**82.7**	**87.4**			**94.0**	**96.0**	**95.8**	**94.9**	**93.6**	**92.4**		**78.3**	**53.8**	
of which: public and publicly guaranteed (PPG)	49.6	52.6	54.8			58.2	59.7	59.4	58.7	57.9	56.9		42.9	15.6	
Change in external debt	6.2	5.9	4.7			6.6	2.0	-0.3	-0.8	-1.3	-1.3		-2.6	-3.2	
Identified net debt-creating flows	10.0	12.6	4.4			4.4	12.7	7.9	10.1	7.5	6.0		3.8	4.3	
Non-interest current account deficit	**12.3**	**17.1**	**11.8**	**16.1**	**4.7**	**11.6**	**12.7**	**14.6**	**14.6**	**11.9**	**10.5**		**8.8**	**9.8**	8.8
Deficit in balance of goods and services	16.6	15.9	13.4			14.3	16.0	17.0	15.9	13.7	11.9		10.3	10.5	
Exports	35.6	33.8	34.3			32.4	31.5	32.6	32.9	33.2	33.5		33.6	33.0	
Imports	52.2	49.8	47.8			46.6	47.6	49.6	48.8	46.9	45.4		43.9	43.5	
Net current transfers (negative = inflow)	-3.4	-3.7	-4.0	-4.3	0.7	-3.9	-3.8	-3.8	-4.0	-4.0	-3.9		-3.8	-3.6	-3.8
of which: official	0.2	-0.6	-1.1			-0.7	-0.4	-0.5	-0.6	-0.6	-0.5		-0.4	-0.3	
Other current account flows (negative = net inflow)	-0.9	4.9	2.3			1.2	0.5	1.4	2.7	2.1	2.4		2.3	2.9	
Net FDI (negative = inflow)	**-4.4**	**-5.3**	**-5.6**	**-6.8**	**2.4**	**-5.5**	**-4.7**	**-4.8**	**-4.8**	**-4.8**	**-4.8**		**-4.9**	**-4.8**	-4.9
Endogenous debt dynamics 2/	**2.1**	**0.9**	**-1.8**			**-1.7**	**4.7**	**-1.8**	**0.3**	**0.4**	**0.3**		**-0.1**	**-0.7**	
Contribution from nominal interest rate	1.4	1.3	1.3			1.7	1.8	1.8	2.0	2.1	2.0		1.3	0.3	
Contribution from real GDP growth	0.0	1.0	-0.5			-3.4	2.8	-3.6	-1.7	-1.7	-1.7		-1.4	-1.0	
Contribution from price and exchange rate changes	0.6	-1.4	-2.7			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-3.9**	**-6.7**	**0.3**	**-7.3**	**6.8**	**2.2**	**-10.6**	**-8.2**	**-10.9**	**-8.8**	**-7.2**		**-6.4**	**-7.5**	
of which: Capital transfers	3.6	2.4	2.6			4.4	6.8	9.4	9.4	7.3	5.2		3.8	6.6	
of which: Commercial banks and other private flows	6.4	2.1	0.8			1.2	0.8	0.4	0.1	0.4	1.1		3.2	0.8	
of which: exceptional financing	0.0	0.0	0.0			0.0	1.7	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	49.8			55.3	59.5	61.1	62.0	62.3	63.2		61.6	56.5	
In percent of exports	...	...	145.1			170.9	188.8	187.4	188.5	188.0	189.0		183.2	171.1	
PV of PPG external debt	**...**	**...**	**17.2**			**19.5**	**23.2**	**24.7**	**25.8**	**26.6**	**27.7**		**26.1**	**18.3**	
In percent of exports	**...**	**...**	**50.1**			**60.3**	**73.6**	**75.9**	**78.5**	**80.2**	**82.9**		**77.6**	**55.3**	
In percent of government revenues	**...**	**...**	**48.2**			**55.9**	**66.4**	**71.9**	**71.9**	**74.3**	**77.8**		**74.2**	**53.2**	
Debt service-to-exports ratio (in percent)	**12.0**	**13.7**	**11.5**			**5.8**	**8.3**	**8.5**	**10.1**	**11.0**	**11.1**		**10.1**	**6.1**	
PPG debt service-to-exports ratio (in percent)	**10.4**	**12.1**	**9.9**			**4.0**	**6.5**	**6.8**	**8.4**	**9.3**	**9.5**		**8.8**	**5.2**	
PPG debt service-to-revenue ratio (in percent)	**10.5**	**11.2**	**9.5**			**3.7**	**5.8**	**6.5**	**7.7**	**8.7**	**8.9**		**8.5**	**5.0**	
Total gross financing need (Billions of U.S. dollars)	0.1	0.1	0.1			0.0	0.1	0.1	0.1	0.1	0.1		0.1	0.1	
Non-interest current account deficit that stabilizes debt ratio	6.2	11.1	7.1			5.0	10.7	14.9	15.5	13.2	11.7		11.5	13.0	
Key macroeconomic assumptions															
Real GDP growth (in percent)	-0.1	-1.3	0.6	2.1	3.1	3.9	-3.0	3.9	1.9	1.9	1.9	1.7	1.9	1.8	1.9
GDP deflator in US dollar terms (change in percent)	-0.8	1.8	3.3	1.7	1.7	-4.0	3.1	-0.1	1.7	2.0	2.0	0.8	2.0	1.3	1.8
Effective interest rate (percent) 5/	2.0	1.7	1.7	2.7	0.9	2.0	2.0	1.9	2.2	2.3	2.2	2.1	1.7	0.6	1.8
Growth of exports of G&S (US dollar terms, in percent)	-3.8	-4.4	5.5	3.7	5.3	-6.1	-2.5	7.4	4.4	4.7	5.0	2.1	3.9	1.6	3.6
Growth of imports of G&S (US dollar terms, in percent)	-2.8	-4.1	-0.3	4.3	8.1	-2.7	2.1	8.3	1.8	-0.2	0.7	1.7	3.9	1.9	3.4
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	45.4	33.5	37.0	35.5	37.5	38.0	37.8	39.5	33.2	38.8
Government revenues (excluding grants, in percent of GDP)	35.3	36.8	35.7			34.9	35.0	34.4	35.9	35.8	35.7		35.1	34.3	34.9
Aid flows (in Billions of US dollars) 7/	0.0	0.1	0.0			0.0	0.1	0.1	0.1	0.1	0.1		0.0	0.0	
of which: Grants	0.0	0.0	0.0			0.0	0.0	0.0	0.1	0.0	0.0		0.0	0.0	
of which: Concessional loans	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			4.2	8.6	11.2	11.5	8.2	6.8		4.6	3.0	4.2
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			61.9	64.3	76.2	76.8	69.7	62.3		65.5	98.7	73.9
Memorandum items:															
Nominal GDP (Billions of US dollars)	0.5	0.5	0.5			0.5	0.5	0.5	0.6	0.6	0.6		0.7	1.0	
Nominal dollar GDP growth	-0.9	0.5	3.9			-0.3	0.0	3.8	3.6	3.8	4.0	2.5	3.9	3.2	3.7
PV of PPG external debt (in Billions of US dollars)			0.1			0.1	0.1	0.1	0.1	0.2	0.2		0.2	0.2	
(PVt-PVt-1)/GDPt-1 (in percent)						2.2	3.7	2.5	2.0	1.8	2.3	2.4	0.7	-0.8	0.3
Gross workers' remittances (Billions of US dollars)	...	...	...			...	...	...	...	...	...		...	...	
PV of PPG external debt (in percent of GDP + remittances)	...	...	17.2			19.5	23.2	24.7	25.8	26.6	27.7		26.1	18.3	
PV of PPG external debt (in percent of exports + remittances)	...	...	50.1			60.3	73.6	75.9	78.5	80.2	82.9		77.6	55.3	
Debt service of PPG external debt (in percent of exports + remittanc	...	...	9.9			4.0	6.5	6.8	8.4	9.3	9.5		8.8	5.2	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table A2. Dominica: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, Baseline Scenario, 2014-34

(In percent)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to GDP ratio								
Baseline	19	23	25	26	27	28	**26**	18
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	19	23	25	25	28	32	**53**	87
A2. New public sector loans on less favorable terms in 2014-2034 2	19	25	28	31	34	37	**43**	43
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	19	23	25	27	27	29	**27**	19
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	19	23	26	28	28	29	**28**	20
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	19	24	25	27	27	29	**27**	19
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	19	24	26	27	28	29	**27**	19
B5. Combination of B1-B4 using one-half standard deviation shocks	19	21	23	24	25	26	**24**	17
B6. One-time 30 percent nominal depreciation relative to the baseline scenario in 2015 5/	19	33	35	36	37	39	**37**	26
PV of debt-to-exports ratio								
Baseline	60	74	76	78	80	83	**78**	55
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	60	74	75	76	84	95	**158**	262
A2. New public sector loans on less favorable terms in 2014-2034 2	60	79	87	94	101	110	**127**	129
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	60	74	76	78	80	83	**78**	55
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	60	72	88	90	92	95	**89**	64
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	60	74	76	78	80	83	**78**	55
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	60	75	79	82	83	86	**81**	58
B5. Combination of B1-B4 using one-half standard deviation shocks	60	65	72	75	76	79	**74**	53
B6. One-time 30 percent nominal depreciation relative to the baseline scenario in 2015 5/	60	74	76	78	80	83	**78**	55
PV of debt-to-revenue ratio								
Baseline	56	66	72	72	74	78	**74**	53
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	56	67	71	70	77	89	**151**	252
A2. New public sector loans on less favorable terms in 2014-2034 2	56	71	82	86	94	103	**122**	124
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	56	65	74	74	76	80	**76**	55
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	56	66	77	77	79	83	**79**	57
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	56	68	74	74	77	80	**76**	55
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	56	68	75	75	77	81	**77**	55
B5. Combination of B1-B4 using one-half standard deviation shocks	56	60	67	67	70	73	**69**	50
B6. One-time 30 percent nominal depreciation relative to the baseline scenario in 2015 5/	56	94	101	101	105	110	**105**	75

Table A2. Dominica: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, Baseline Scenario, 2014-34 (concluded)

(In percent)

Debt service-to-exports ratio								
Baseline	4	6	7	8	9	10	**9**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	4	6	7	8	9	9	**11**	16
A2. New public sector loans on less favorable terms in 2014-2034 2	4	6	6	7	9	9	**10**	12
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	4	6	7	8	9	10	**9**	5
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	4	6	7	9	10	11	**10**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	4	6	7	8	9	10	**9**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	4	6	7	9	9	10	**9**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	4	6	7	8	9	10	**9**	5
B6. One-time 30 percent nominal depreciation relative to the baseline scenario in 2015 5/	4	6	7	8	9	10	**9**	5
Debt service-to-revenue ratio								
Baseline	4	6	6	8	9	9	**8**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	4	6	6	7	8	9	**11**	15
A2. New public sector loans on less favorable terms in 2014-2034 2	4	6	6	7	8	8	**9**	11
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	4	6	7	8	9	9	**9**	5
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	4	6	6	8	9	9	**9**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	4	6	7	8	9	9	**9**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	4	6	7	8	9	9	**9**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	4	6	6	8	9	9	**8**	5
B6. One-time 30 percent nominal depreciation relative to the baseline scenario in 2015 5/	4	8	9	11	12	13	**12**	7
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	19	19	19	19	19	19	**19**	19

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline scenario, while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline scenario level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table A3. Dominica: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011-34

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	69.7	72.6	74.7			79.8	82.7	82.7	80.5	79.7	78.8		59.8	21.2	
of which: foreign-currency denominated	49.6	52.6	54.8			58.2	59.7	59.4	58.7	57.9	56.9		42.9	15.6	
Change in public sector debt	2.9	2.8	2.2			5.1	2.9	0.0	-2.2	-0.8	-1.0		-4.4	-3.5	
Identified debt-creating flows	5.5	5.5	0.3			5.1	2.9	0.0	-2.2	-0.8	-1.0		-4.4	-3.6	
Primary deficit	3.0	4.2	1.1	-1.5	4.2	2.8	0.2	0.3	-2.4	-1.0	-1.0	-0.2	-4.2	-3.4	-4.1
Revenue and grants	40.1	39.5	38.9			36.9	41.2	43.6	45.5	41.9	39.9		38.1	37.3	
of which: grants	4.8	2.7	3.2			2.1	6.2	9.1	9.6	6.1	4.3		3.0	3.0	
Primary (noninterest) expenditure	43.1	43.6	40.0			39.8	41.4	43.9	43.1	40.9	38.9		33.9	34.0	
Automatic debt dynamics	2.5	1.4	-0.8			2.2	2.7	-0.3	0.2	0.2	0.1		-0.1	-0.2	
Contribution from interest rate/growth differential	1.3	1.4	0.1			-1.1	3.2	-1.6	0.0	0.2	0.1		-0.2	-0.3	
of which: contribution from average real interest rate	1.2	0.5	0.5			1.7	0.7	1.6	1.5	1.6	1.6		1.0	0.2	
of which: contribution from real GDP growth	0.0	0.9	-0.4			-2.8	2.5	-3.1	-1.5	-1.5	-1.5		-1.2	-0.5	
Contribution from real exchange rate depreciation	1.3	0.0	-0.9			3.3	-0.5	1.3	0.2	0.0	0.0		0.0	0.1	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-2.6	-2.7	1.9			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	37.2			41.1	46.2	48.1	47.6	48.4	49.6		43.0	23.9	
of which: foreign-currency denominated	...	...	17.2			19.5	23.2	24.7	25.8	26.6	27.7		26.1	18.3	
of which: external	...	...	17.2			19.5	23.2	24.7	25.8	26.6	27.7		26.1	18.3	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	10.1	11.0	6.6			5.4	4.5	5.2	4.9	5.3	6.5		3.6	0.3	
PV of public sector debt-to-revenue and grants ratio (in perce	...	...	95.6			111.4	112.1	110.4	104.5	115.6	124.3		112.8	64.1	
PV of public sector debt-to-revenue ratio (in percent)	...	...	104.1			118.1	132.0	139.7	132.5	135.3	139.1		122.5	69.7	
of which: external 3/	...	...	48.2			55.9	66.4	71.9	71.9	74.3	77.8		74.2	53.2	
Debt service-to-revenue and grants ratio (in percent) 4/	17.8	17.4	14.1			7.1	10.5	11.2	16.2	14.9	18.8		20.6	9.8	
Debt service-to-revenue ratio (in percent) 4/	20.3	18.6	15.3			7.5	12.4	14.2	20.6	17.5	21.1		22.3	10.7	
Primary deficit that stabilizes the debt-to-GDP ratio	0.0	1.3	-1.0			-2.2	-2.7	0.3	-0.2	-0.2	-0.1		0.1	0.2	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	-0.1	-1.3	0.6	2.1	3.1	3.9	-3.0	3.9	1.9	1.9	1.9	1.7	1.9	1.8	1.9
Average nominal interest rate on forex debt (in percent)	1.8	1.6	1.6	2.6	1.0	2.1	2.2	2.2	2.6	2.8	2.7	2.4	2.0	0.2	2.5
Average real interest rate on domestic debt (in percent)	6.8	2.9	2.5	3.3	2.5	8.7	3.3	7.0	5.4	5.4	5.6	5.9	5.6	7.7	6.2
Real exchange rate depreciation (in percent, + indicates depr	2.6	-0.1	-1.7	0.3	1.8	6.3	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	-0.8	1.8	3.3	1.7	1.7	-4.0	3.1	-0.1	1.7	2.0	2.0	0.8	2.0	1.3	1.8
Growth of real primary spending (deflated by GDP deflator, ir	-11.6	0.0	-7.8	-1.9	4.2	3.2	1.0	10.2	0.1	-3.3	-3.1	1.3	2.6	3.0	1.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	45.4	33.5	37.0	35.5	37.5	38.0	37.8	39.5	33.2	...

Sources: Country authorities; and staff estimates and projections.

1/ Public and publicly guaranteed gross debt.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

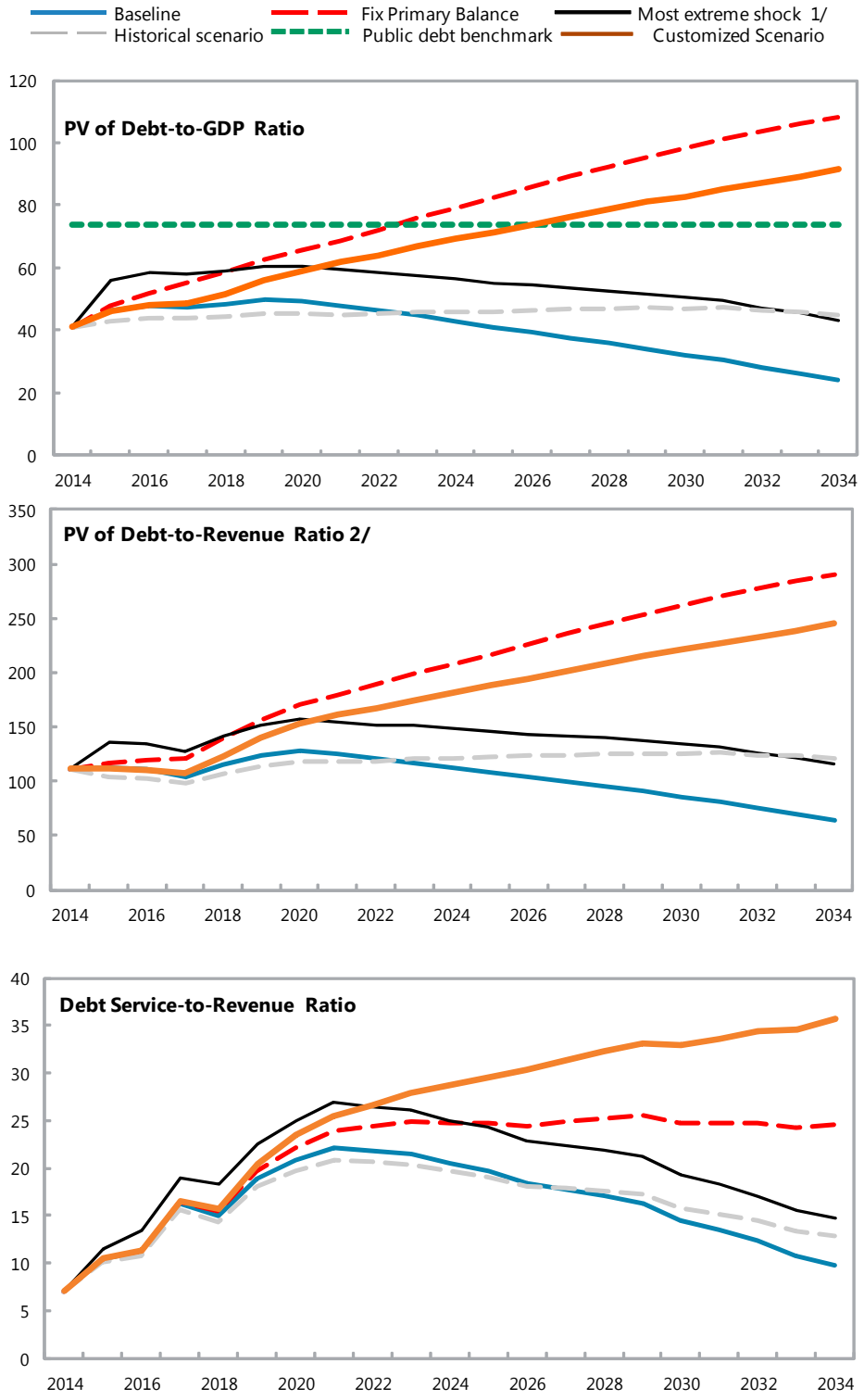
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table A4. Dominica: Sensitivity Analysis for Key Indicators of Public Debt, Baseline Scenario, 2014-34

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline Scenario	41	46	48	48	48	50	43	24
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	41	43	44	44	44	45	46	45
A2. Primary balance is unchanged from 2014	41	48	52	55	59	63	79	108
A3. Permanently lower GDP growth 1/	41	47	49	50	51	54	55	64
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	41	45	50	50	51	53	50	37
B2. Primary balance is at historical average minus one standard deviations in 2015-201	41	48	52	51	52	53	46	27
B3. Combination of B1-B2 using one half standard deviation shocks	41	45	48	48	49	50	43	23
B4. One-time 30 percent real depreciation in 2015	41	56	58	58	59	61	57	43
B5. 10 percent of GDP increase in other debt-creating flows in 2015	41	53	55	55	56	57	50	30
PV of Debt-to-Revenue Ratio 2/								
Baseline Scenario	111	112	110	105	116	124	113	64
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	111	104	102	98	107	114	121	121
A2. Primary balance is unchanged from 2014	111	117	119	121	140	157	208	290
A3. Permanently lower GDP growth 1/	111	113	113	108	122	134	145	170
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	111	109	113	109	122	133	131	99
B2. Primary balance is at historical average minus one standard deviations in 2015-201	111	116	118	112	124	133	122	72
B3. Combination of B1-B2 using one half standard deviation shocks	111	109	111	105	116	125	112	62
B4. One-time 30 percent real depreciation in 2015	111	137	134	127	141	152	148	116
B5. 10 percent of GDP increase in other debt-creating flows in 2015	111	129	127	120	133	142	131	81
Debt Service-to-Revenue Ratio 2/								
Baseline Scenario	7	10	11	16	15	19	21	10
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	7	10	11	16	14	18	20	13
A2. Primary balance is unchanged from 2014	7	10	11	17	16	20	25	25
A3. Permanently lower GDP growth 1/	7	11	11	17	15	20	23	17
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	7	10	11	17	15	19	22	12
B2. Primary balance is at historical average minus one standard deviations in 2015-201	7	10	11	17	15	19	21	11
B3. Combination of B1-B2 using one half standard deviation shocks	7	10	11	16	15	19	21	10
B4. One-time 30 percent real depreciation in 2015	7	12	13	19	18	22	25	15
B5. 10 percent of GDP increase in other debt-creating flows in 2015	7	10	12	17	15	19	22	11

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Figure A2. Dominica: Indicators of Public Debt Under Alternative Scenarios, 2014-34 (Baseline Scenario) 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.